



# Active Transportation & the Built Environment

Accessibility, Design, and Planning

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**Would you walk one mile?**

Would you walk one mile  
to grab a coffee?



Would you walk one mile  
to get to a BART station?

Would you walk one mile  
for a medical appointment?

# Trip Purpose

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Walking-distance distributions change substantially with trip purpose as it can affect both the willingness to walk or roll and the distance people are willing to travel.

In Yang & Diez-Roux (2012):

- How did trip purpose affect walking distance?
- For which types of trips did people walk longer distances?

# Distance is an outcome

Yang & Diez-Roux (2012)

Shows that observed walking distance varies when trips are stratified by purpose and population subgroup



Would you walk one mile  
to a transit station in Berkeley,  
at 11 pm, after a long day,  
carrying a heavy bag?

**with a stroller**

**95°F heat**

**with  
two grocery bags**

**Would you walk  
one mile?**

**50-step staircase  
twice**

**gravel walkway**

**No sidewalk for half  
of the way**

**with a stroller**

**95°F heat**

**with  
two grocery bags**

If we estimate one “acceptable  
walking distance,” whose trips  
produced that number?

**50-step staircase  
twice**

**gravel walkway**

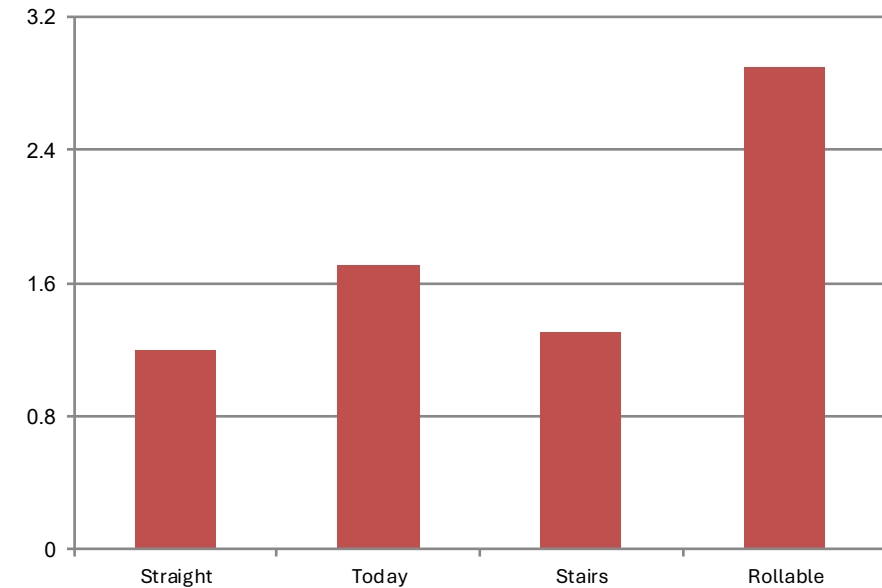
**No sidewalk for half  
of the way**

***Distance describes space. It does not, by itself, describe accessibility.***

| Measure                   | Question                                                                 | Assumption                          |
|---------------------------|--------------------------------------------------------------------------|-------------------------------------|
| Straight line distance    | How far apart are origin and destination?                                | space between points is traversable |
| Network distance          | How far must someone travel using the available street/path network?     | every mapped edge is usable         |
| Feasible network distance | How far must this person travel using routes they can actually traverse? | edge rules are traveler specific    |

# The same OD pair has multiple defensible distances

|               |                                                  |
|---------------|--------------------------------------------------|
| <b>1.2 mi</b> | straight-line distance                           |
| <b>1.7 mi</b> | today: continuously accessible network route     |
| <b>1.3 mi</b> | 2001: stair route, feasible only if stairs count |
| <b>2.9 mi</b> | 2001: continuously rollable / rideable route     |



Approximate lengths from the classroom route. Treat them as a measurement argument, not official engineering measurements.

# Is this road a barrier?

For whom?

Trying to reach what?

By which route and mode?

Under which crossing conditions?



# Infrastructure as Barriers

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The same infrastructure can enable one trip, constrain another, and generate different consequences for different people.

Verdict: Bike lanes and fixed buffers can create emergency response time longer

Verdict: tactile paving is a nightmare for people using wheelchair, walker, cane

# Readings at a Glance



## Who is walking?

Yang & Diez-Roux (2012)

Observed walking distance varies when trips are stratified by purpose and population subgroup.

## At what scale?

van Eldijk, Gil & Marcus (2022)

The same facility can connect the city while acting as a barrier for specific users and trips. Connection and severance are measured at different scales.

## To what end?

Mueller et al. (2015)

The benefits of mode shift arrive through separate pathways: safety, physical activity, pollution, emissions. One score cannot stand in for all of them.

**walkable for whom, at what scale, toward which outcome?**

Large data can be used well if we accept what it is: A description of the people it captured, not a description of the population

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# Takeaways

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1. State the unit and the population clearly early on.
2. One relationship rarely holds for everyone, so stratify by theoretically meaningful groups or contexts.
3. Design for the constraint, not the average.
4. Report who the instrument cannot see.
5. Report the aggregate and the distribution.

# Five questions to carry forward

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1. For whom?
2. For what trip?
3. What enables or constrains it?
4. What do people do in response?
5. What consequences follow and compared with what?